

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 3 – Hackathon**

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5-AT3 Hackathon	Assessment:	ASSESSMENT TOOL 3 – Hackathon
Weightage:	30% of CIA	Total Marks:	100
CO4 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	V.Naga Nischay	Reg. No.:	192472255
Batch	2024	Date:	29-08-2026

Code of Conduct

I V.Naga Nischay (192472255) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.Naga Nischay

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

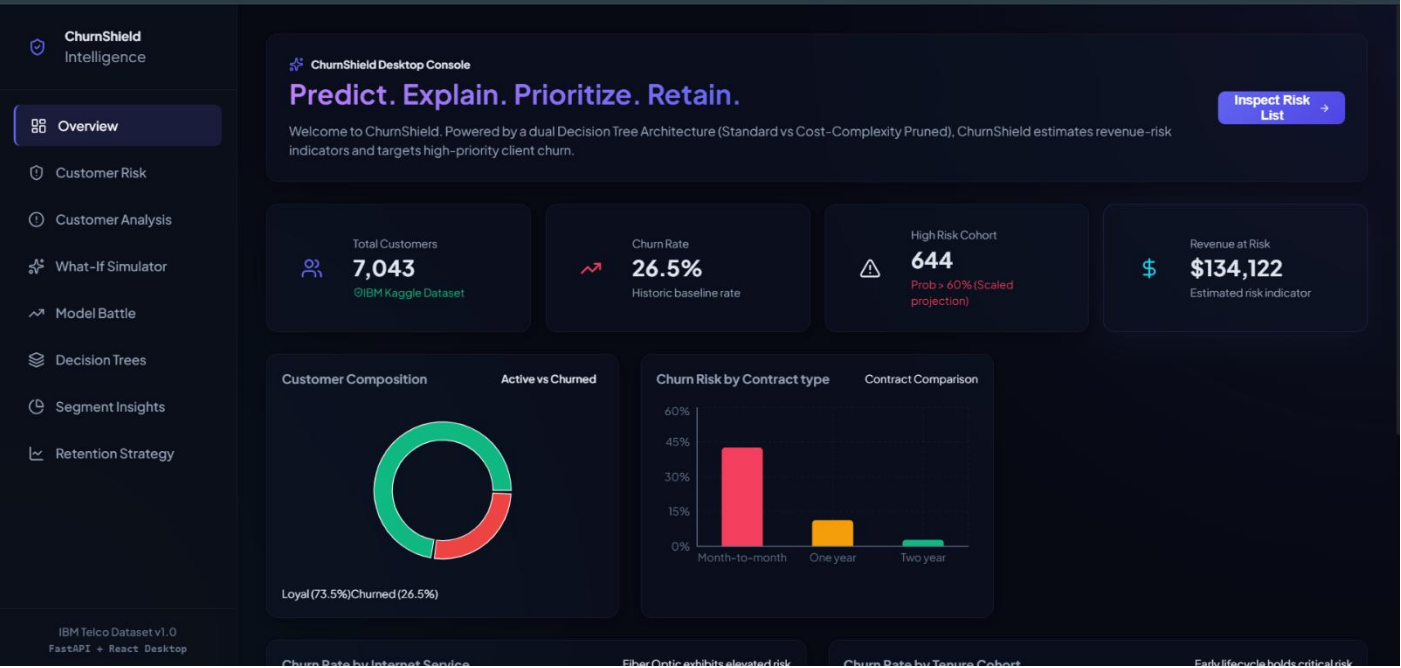
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Problem Statement:

Students at academic risk are often identified too late for effective intervention. This system uses attendance, past grades, and study hours with the kNN algorithm to classify students as At Risk or Safe, helping educators identify at-risk students early and provide timely academic support.

Code:

```
... download_data.py main.py X README.md \ README.md frontend App.js RetentionStrategy.js
backend > app > main.py
1 import os
2 import shutil
3 import joblib
4 import numpy as np
5 import pandas as pd
6 from typing import Dict, List, Optional
7 from fastapi import FastAPI, UploadFile, File, HTTPException, Query
8 from fastapi.middleware.cors import CORSMiddleware
9 from pydantic import BaseModel
10 from sklearn.metrics import precision_recall_fscore_support
11
12 from .data_manager import TelcoDataManager, CAT_COLS, NUM_COLS, CSV_PATH, MODELS_DIR
13 from .model import TelcoModelTrainer
14
15 app = FastAPI(title="ChurnShield AI Backend API")
16
17 # Enable CORS for frontend integration
18 app.add_middleware(
19     CORSMiddleware,
20     allow_origins=["*"], # For hackathon/local development
21     allow_credentials=True,
22     allow_methods=["*"],
23     allow_headers=["*"],
24 )
25
26 # Global variables to hold data and models
27 data_manager = TelcoDataManager()
28 trainer = None
29 standard_model = None
30 pruned_model = None
31 selected_ccp_alpha = 0.0
32
33 class CustomerInput(BaseModel):
34     gender: str = "Male"
35     SeniorCitizen: int = 0
36     Partner: str = "No"
37     Dependents: str = "No"
38     tenure: int = 1
39     PhoneService: str = "No"
40     MultipleLines: str = "No"
41     InternetService: str = "No"
42     OnlineSecurity: str = "No"
43     OnlineBackup: str = "No"
44     DeviceProtection: str = "No"
45     TechSupport: str = "No"
46     StreamingTV: str = "No"
47     StreamingMovies: str = "No"
48     Contract: str = "Month-to-month"
49     PaperlessBilling: str = "No"
50     PaymentMethod: str = "Electronic check"
51     MonthlyCharges: float = 50.0
52     TotalCharges: str = "" # Let it calculate automatically if empty or raw string
53
54 def initialize_system():
55     global data_manager, trainer, standard_model, pruned_model, selected_ccp_alpha
56
57     # 1. Download dataset if not exists
58     from .download_data import download_dataset
```



ChurnShield
Intelligence

Overview

Customer Risk

Customer Analysis

What-If Simulator

Model Battle

Decision Trees

Segment Insights

Retention Strategy

IBM Telco Dataset v1.0
FastAPI + React Desktop

All predictions

Matches: 1,409

CUSTOMER ID	CHURN PROB.	RISK LEVEL	MONTHLY CHARGE	PRIORITY SCORE	REV. AT RISK	PREDICTED CHURN	RECOMMENDED ACTION	DETAILS
7181-BQYBV	85.5%	HIGH	\$102.45	87.6	\$87.59	Yes	Offer Annual Discount	
6857-VMJDT	85.5%	HIGH	\$95.65	81.8	\$81.77	Yes	Offer Annual Discount	
0295-PPHDO	85.5%	HIGH	\$95.45	81.6	\$81.60	Yes	Offer Annual Discount	
5178-LMXOP	85.5%	HIGH	\$95.10	81.3	\$81.30	Yes	Offer Annual Discount	
3292-PBZEF	72.1%	HIGH	\$111.40	80.4	\$80.36	Yes	Offer Annual Discount	
9851-KIELU	72.1%	HIGH	\$110.10	79.4	\$79.43	Yes	Offer Annual Discount	
5167-ZFFMM	85.5%	HIGH	\$90.85	77.7	\$77.67	Yes	Offer Annual Discount	
2027-FECZV	72.1%	HIGH	\$106.70	77.0	\$76.97	Yes	Offer Annual Discount	
0970-ETWGE	85.5%	HIGH	\$89.55	76.6	\$76.56	Yes	Offer Annual Discount	

